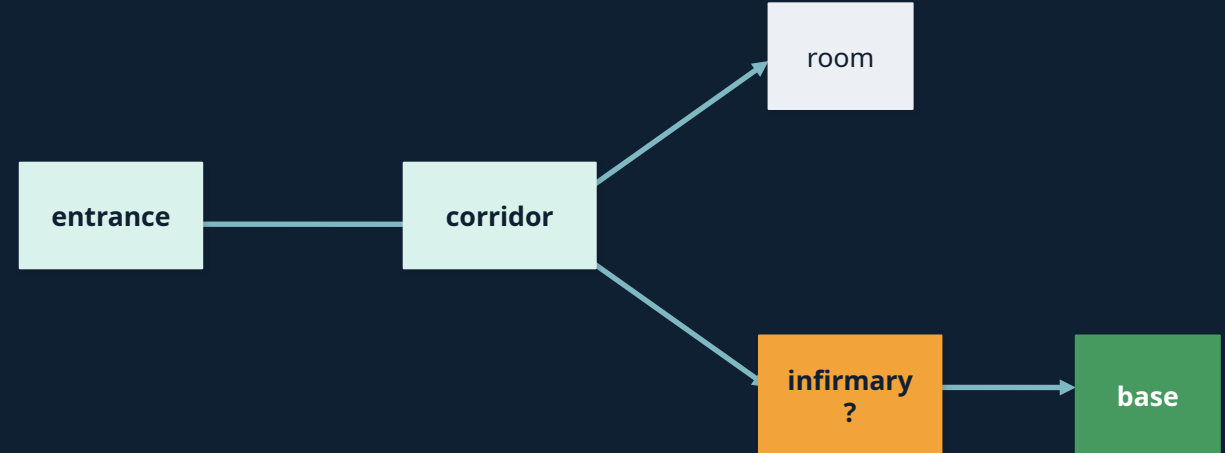


D4-V2 Search and Rescue

Unknown Victim Location

A rescue robot must explore a known topology, inspect rooms, detect the victim, and complete rescue before time-dependent health degradation makes the mission fail.



FINAL VERSION TO CHECK
codes/D4-V2_search_and_rescue_definitive

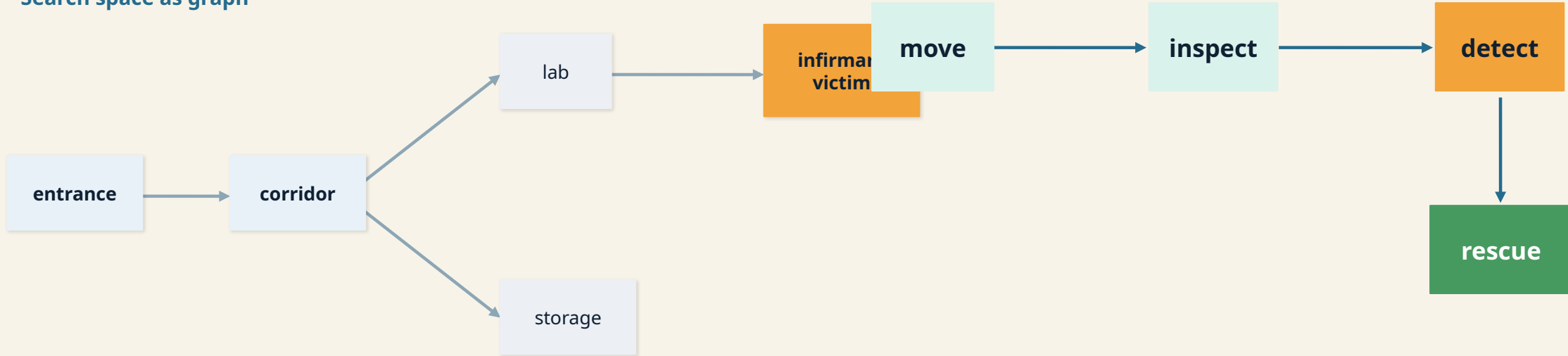
6 slides · 10 minute presentation

Problem overview → modelling choices → final demo → limitations

Q1 — Classical PDDL approximation

Unknown victim location is approximated through explicit inspection predicates.

Search space as graph



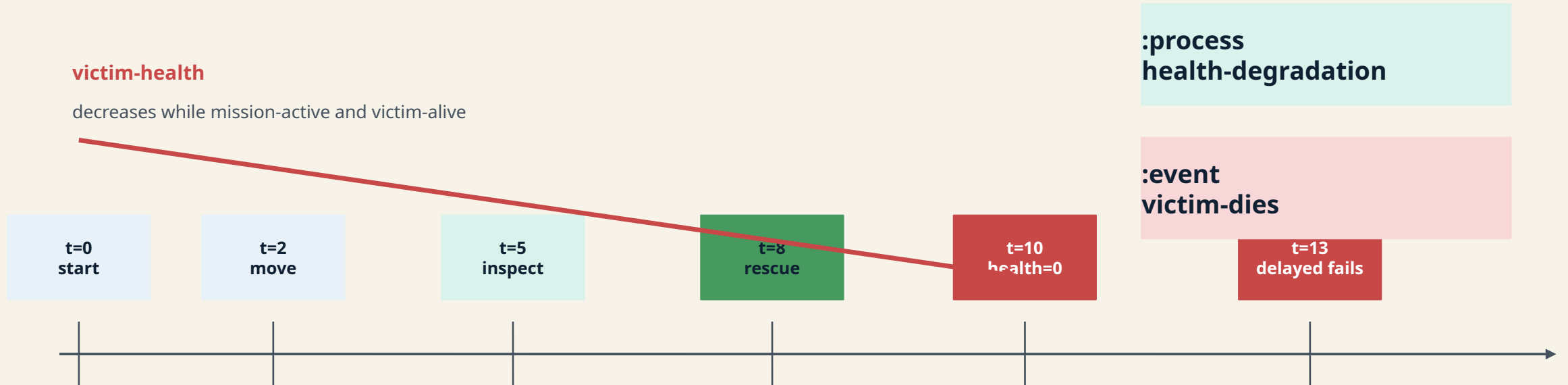
The planner cannot use hidden uncertainty directly.
Rooms become inspectable symbolic states.
rescue depends on victim-detected, so direct rescue is blocked.

Key answer for the presentation

Classical PDDL gives a valid abstraction, not real partial observability.
The model makes information acquisition explicit through inspection actions.

Q2 — PDDL+ adds time pressure

The world changes between robot actions: victim health decreases continuously.



Fast rescue succeeds at t=8
Delayed rescue fails before t=13

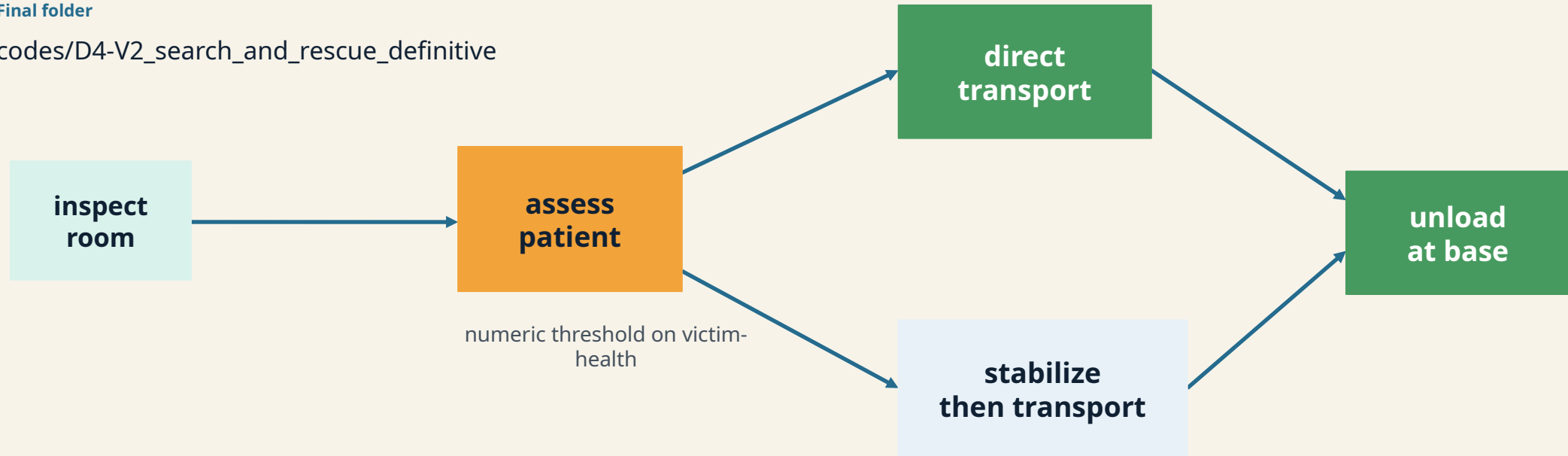
Inactivity is no longer neutral.
Events are automatic threshold effects.
The planner must reason over trajectories, not only symbolic states.

Final definitive strategy — autonomous branch choice

The final version chooses direct transport or stabilization from numeric health, not from manual branch facts.

Final folder

codes/D4-V2_search_and_rescue_definitive



**mission-clock
+ mission-timeout**

Why added?

It prevents the planner from waiting artificially until health crosses a threshold just to force another branch.

Validation and demo outputs

The live demo can be replaced by saved planner outputs if the planner environment fails.

Direct transport Solved

Plan contains transport actions and does not contain start-stabilize-patient.

Found Plan:
15.0 start-move
21.0 start-inspect
22.0 start-assess
23.0 start-load
33.0 start-unload

Stabilize + transport Solved

Plan contains start-stabilize-patient before loading and moving the patient.

Found Plan:
10.0 start-assess
11.0 start-stabilize
15.0 start-load
29.0 start-unload

Too-late case Unsolvable

Forced delay makes the mission fail before rescue can be achieved.

Problem unsolvable
Metric: -1
Dead-Ends detected

Demo path: open final domain + one problem in VS Code, run ENHSP local Docker, compare with saved plan file.
Fallback: read the three saved outputs in the final folder.

Reflection, limitations, and final delivery

The model is deliberately symbolic: it makes planning tractable but abstracts away real execution.

What works

Q1 enforces inspect → detect → rescue.
Q2 shows health/time interaction.
Final version validates direct, stabilization, and failure cases.

Known limits

No real partial observability or belief state.
No noisy sensing, geometry, motion planning, or execution error.
Some delayed PDDL+ planner outputs are documented as unsupported.

Repository

FINAL VERSION TO CHECK:
codes/D4-V2_search_and_rescue_definitive

Report/report.pdf
slide/presentation.pdf

Planning abstraction

PDDL+ dynamics

Validated outputs

Clear limits

Closing message: the assignment demonstrates how a planning model can encode information acquisition, time pressure, and branch choice, while making its abstractions explicit.